

GROWcottage

CONTAINERIZED PLANT PRODUCTION SYSTEM

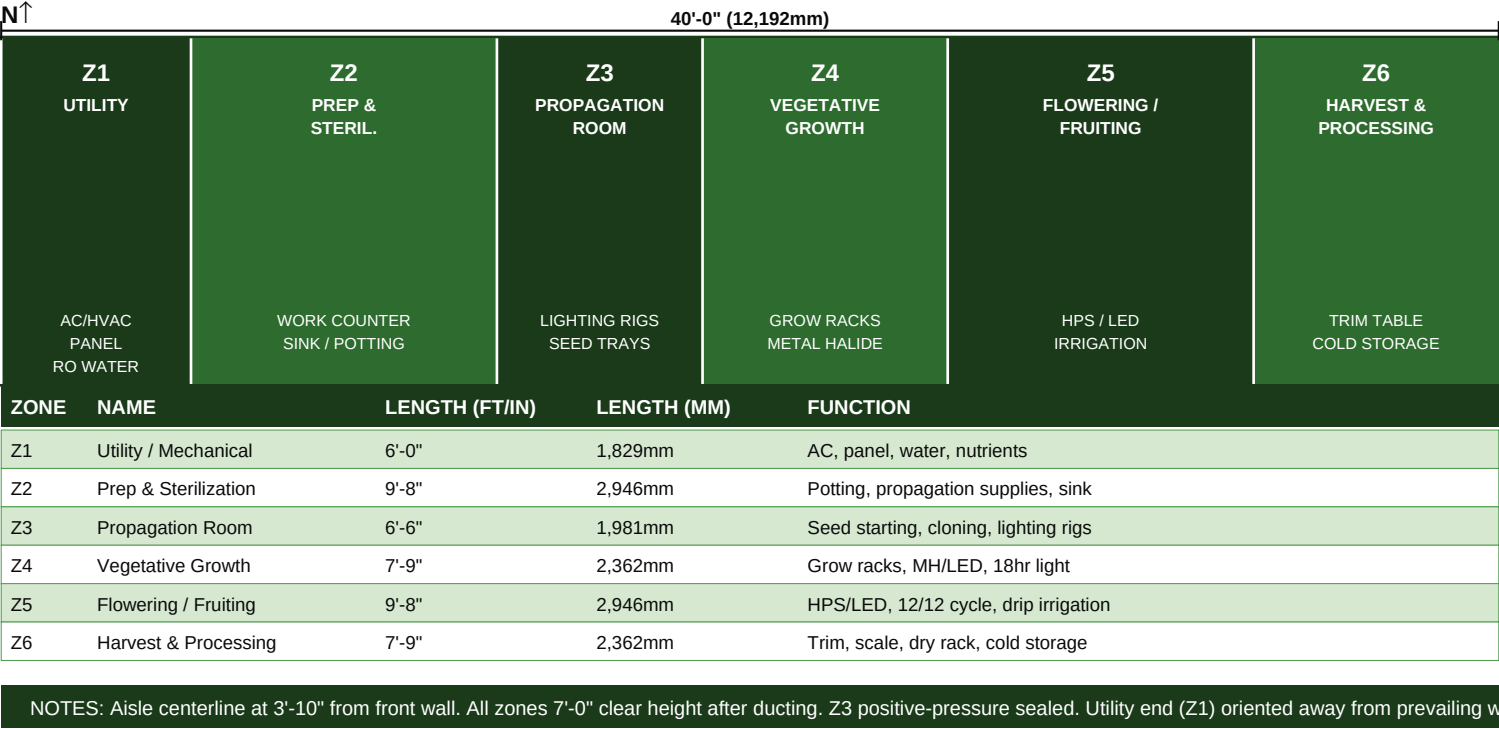
CONTAINER	PRODUCTION	OUTPUT TARGET	INTEGRATION	INSTALL TYPE
40FT HIGH CUBE 9'6" INTERIOR HT	ALL STAGES FULL CYCLE	20–100 LBS/WK COMMERCIAL	GRUSH BATCH TRACKING	SITE BUILT ZONE 9B

This document is the official build specification for the GROWcottage containerized plant production system. It contains dimensioned floor plans, zone specifications, electrical single-line diagram, equipment and parts list, environmental targets, and a phased build checklist. This plan package is licensed for single-use construction. Redistribution or resale requires a contractor license from Grush.

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01 — DIMENSIONED FLOOR PLAN

40ft High Cube container interior: 39'5" × 7'8" × 8'10" usable (11,989mm × 2,337mm × 2,692mm). All dimensions to finished wall face after insulation and paneling. Zones separated by removable partition systems unless noted as fixed.



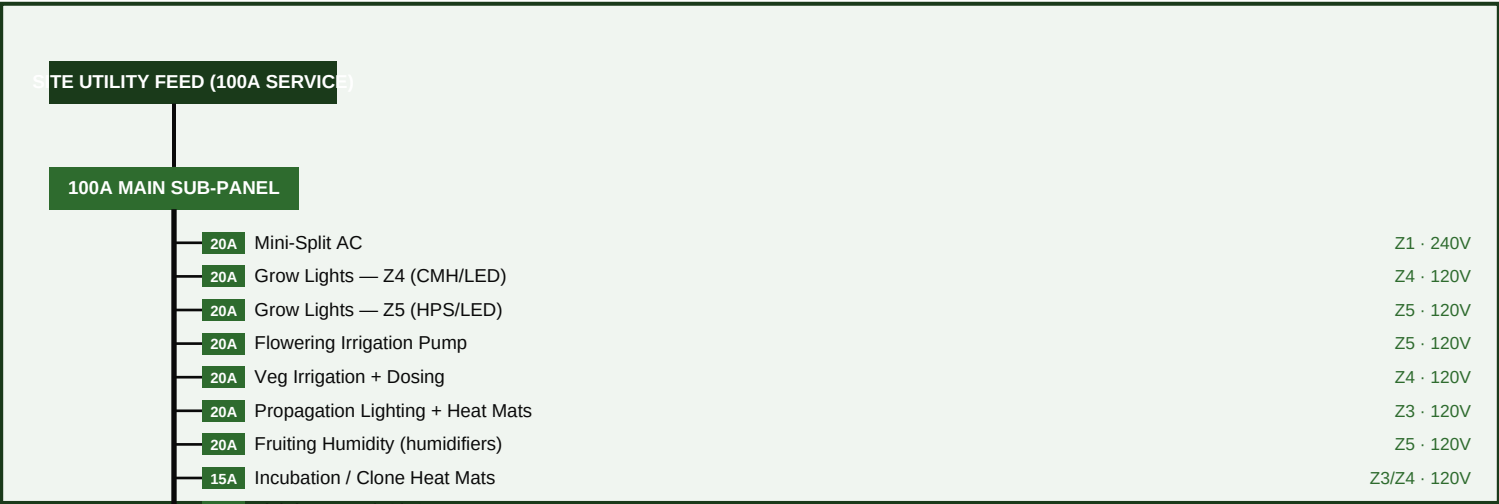
02 — ZONE SPECIFICATIONS

Z#	ZONE NAME	DESCRIPTION & KEY EQUIPMENT	PARTITION
Z1	Utility / Mechanical	18,000 BTU mini-split AC unit (wall mount). 100A sub-panel. 5-stage RO water system (75 GPD). Nutrient dosing reservoir (55-gal). CO2 enrichment tank + controller (optional).	Fixed wall (plywood + FRP)
	Prep & Sterilization	Stainless work counter with utility sink (8ft). Potting bench. Propagation supply shelving. Bag cooling racks. Growing media mixing station (coco coir, perlite, amendments).	Open to aisle
	Propagation Room	T5/LED germination lighting rigs x2 (4ft, 4-tube). Heat mats x4 on thermostat. Humidity dome staging shelves. Clone collars and aeroponic cloner (24-site). Positive pressure HEPA supply.	Acrylic panel + door seal
Z4	Vegetative Growth	NSF wire shelving racks x2 (6 shelves each, 74" posts). CMH/LED grow lights — 315W per shelf tier. 18-hour light cycle timer. Oscillating fans x2. Drip irrigation manifold (veg). 65–78°F target.	Removable curtain
Z5	Flowering / Fruiting	NSF wire shelving racks x3 (6 shelves each). HPS 600W or LED equiv per tier. 12/12 light cycle. Drip irrigation system with timer. Ultrasonic humidifiers x2 on Inkbird RH controller. AC Infinity inline fans x2 for FAE. Light-tight blackout curtain at partition.	Blackout curtain
Z6	Harvest & Processing	Harvest/trim table (stainless, 6ft). Drying racks (multi-tier, mesh). Digital scale (0.1g, 30kg). Vacuum sealer. Upright refrigerator (20 cu ft, 34-38°F). Grush batch-close scan point.	Open to aisle

GRUSH INTEGRATION points at Z3 (batch open/propagation), Z4 entry (veg load — shelf location logged), Z5 entry (flower transfer — stage change), and Z6 (harvest weight log — batch close or flush extension). Environmental sensor data feeds into batch records via Home

03 — ELECTRICAL SINGLE-LINE DIAGRAM

100A service minimum. All branch circuits self-performed by licensed wireman. Utility tie-in to site meter by licensed electrician with permit. Container shell bonded to ground.



GROUNDING: Container shell bonded to ground rod. All metal conduit bonded. Utility tie-in by licensed electrician · Branch circuits self-performed.

CIRCUIT	VOLTAGE	EST. LOAD	ZONE	NOTES
Mini-Split AC	20A / 240V	~1,500W	Z1	Dedicated; HVAC installer connects
Grow Lights Z4 (CMH/LED)	20A / 120V	~600W	Z4	Per-shelf timers; 18hr veg cycle
Grow Lights Z5 (HPS/LED)	20A / 120V	~800W	Z5	Per-shelf timers; 12/12 flower cycle
Flower Irrigation Pump	20A / 120V	~200W	Z5	Timer-controlled drip manifold
Veg Irrigation + Dosing	20A / 120V	~150W	Z4	Nutrient dosing pump on timer
Propagation Lighting	20A / 120V	~300W	Z3	T5/LED + heat mats on thermostat
Fruiting Humidity	20A / 120V	~600W	Z5	Humidifiers + Inkbird controller
GFCI Outlets (x12)	20A / 120V	~1,800W	All	Duplex every 4ft along back wall
Cold Storage Fridge	20A / 120V	~150W run	Z6	Dedicated — compressor surge
Sensors / Controls Hub	15A / 120V	~50W	Z1	Raspberry Pi, sensors, Grush tablet

04 — EQUIPMENT & PARTS LIST

Mid-range pricing sourced from public retail listings as of June 2026. Prices are estimates — verify before purchase. Substitute equivalent models where noted. Contact us for more information.

HVAC & CLIMATE					
ITEM	QTY	EST. UNIT \$	EST. TOTAL	ZONE	SOURCE
Mini-split AC, 18,000 BTU ductless	1	\$950	\$950	Z1	Pioneer / MRCOOL
Inkbird humidity controller IHC-200	2	\$60	\$120	Z3/Z5	Amazon
Ultrasonic humidifier, 6L	2	\$90	\$180	Z5	Amazon
AC Infinity CLOUDLINE T6 inline fan	3	\$120	\$360	All	AC Infinity
HEPA filter inline housing	2	\$90	\$180	Z3/Z5	Amazon
Inkbird temp controller ITC-308	2	\$35	\$70	Z4/Z5	Amazon
Oscillating clip fan, 6in	4	\$20	\$80	Z4/Z5	Amazon
ELECTRICAL					
ITEM	QTY	EST. UNIT \$	EST. TOTAL	ZONE	SOURCE
Square D QO 100A sub-panel, 24-space	1	\$180	\$180	Z1	Home Depot
GFCI duplex outlet, 20A (Leviton)	24	\$13	\$312	All	Home Depot
Misc wire, conduit, boxes, hardware	lot	—	\$800	All	Home Depot
GROW LIGHTING					
ITEM	QTY	EST. UNIT \$	EST. TOTAL	ZONE	SOURCE
CMH 315W grow light fixture	4	\$280	\$1,120	Z4	Amazon / HTG Supply
HPS 600W grow light + digital ballast	3	\$220	\$660	Z5	Amazon / HTG Supply
T5 4ft 4-tube germination light	2	\$120	\$240	Z3	Amazon
Smart outlet / light timer	8	\$18	\$144	Z4/Z5	Amazon
IRRIGATION					
ITEM	QTY	EST. UNIT \$	EST. TOTAL	ZONE	SOURCE
Drip irrigation manifold kit (16-zone)	1	\$180	\$180	Z4/Z5	DripWorks
Submersible pump, 400 GPH	2	\$45	\$90	Z4/Z5	Amazon
Nutrient reservoir, 55-gal food-grade	1	\$80	\$80	Z1	Local / Amazon
Inkbird irrigation timer controller	2	\$40	\$80	Z4/Z5	Amazon
Drip emitters, tubing, fittings — lot	1	—	\$120	All	DripWorks
SHELVING & FURNISHINGS					
ITEM	QTY	EST. UNIT \$	EST. TOTAL	ZONE	SOURCE
NSF wire shelf unit, 74in, 6-shelf	5	\$150	\$750	Z2-Z6	Webstaurant
Stainless work counter, 8ft	1	\$320	\$320	Z2	Webstaurant
Stainless harvest/trim table, 6ft	1	\$240	\$240	Z6	Webstaurant
Rolling utility cart, 3-shelf	2	\$80	\$160	All	Amazon
Upright refrigerator, 20 cu ft	1	\$680	\$680	Z6	Home Depot
Mesh drying rack, multi-tier	2	\$60	\$120	Z6	Amazon

COST SUMMARY	
Total Parts (mid-range estimate)	~\$9,190
Sub labor: HVAC install + refrigerant	~\$600
Sub labor: Plumber (supply + drain)	~\$800
Sub labor: Spray foam crew	~\$900
Licensed electrician (utility tie-in only)	~\$400
Contingency (10%)	~\$1,190
TOTAL BUILD ESTIMATE (excl. container)	~\$13,080
Optional: 40HC container (used, shipyard direct) \$1,000–\$2,000	

05 — ENVIRONMENTAL TARGETS

Target ranges for each zone. Sensors log to Home Assistant and export to Grush batch records. Acceptable variance: +/-2 degrees F temperature, +/-5% RH. Light hours are set in the Grush batch records.

ZONE	TEMP TARGET	TEMP (C)	RH %	LIGHT CYCLE	CO2 (PPM)	AIRFLOW
Z1 — Utility	Ambient	Ambient	—	Work light	<1,000	Exhaust fan
Z2 — Prep	65–80°F	18–27°C	40–60%	5000K work	<1,000	Exhaust above bench
Z3 — Propagation	72–78°F	22–26°C	65–80%	18hr / 6off	<800	Positive pressure
Z4 — Vegetative	68–78°F	20–26°C	50–70%	18hr on / 6off	<1,200	Active — oscillating
Z5 — Flowering	65–75°F	18–24°C	45–65%	12hr on / 12off	<1,200	Active FAE timed
Z6 — Processing	60–65°F	15–18°C	40–55%	5000K work	<800	Exhaust only
Z6 — Cold Storage	34–38°F	1–3°C	—	Fridge interior	—	Fridge compressor

SENSOR PLACEMENT GUIDE

SENSOR TYPE	MODEL	ZONES	PLACEMENT	DATA DESTINATION
Temp / RH	SHT31	Z2–Z5	Center of zone, 5ft height, shaded from direct airflow	Home Assistant → Grush
CO2	SCD40	Z4, Z5	Return air path, 4ft height	Home Assistant → Grush
RH Controller	Inkbird IHC-200	Z3, Z5	Probe mid-zone, controller at Z1 panel	Direct relay control
Temp Controller	Inkbird ITC-308	Z4	Probe near rack center	Direct relay control
Light Timer	Smart plug / mech	Z4, Z5	Grow light circuits; separate veg and flower zones	Independent
VPD Monitor	Govee TH Elite	Z4, Z5	Canopy level — track vapor pressure deficit	Home Assistant

VPD TARGET GUIDE — VAPOR PRESSURE DEFICIT BY STAGE

Propagation: 0.4–0.8 kPa | Vegetative: 0.8–1.2 kPa | Flowering early: 1.0–1.5 kPa | Flowering late: 1.2–1.6 kPa
Use temp + RH readings per zone to calculate VPD. Target the lower end of each range during lights-off. High VPD (dry) increases transpiration — monitor for wilting.

06 — BUILD PHASE CHECKLIST

Sequential build order. Complete each phase before beginning the next. DIY = self-performed. SUB = licensed trade required. EITHER = your call.

PHASE 1 — CONTAINER ACQUISITION & PREP

TASK	WHO
☐ Inspect container — no fumigation residue, no structural damage	DIY
☐ Pressure wash interior, remove all debris	DIY
☐ Apply rust treatment to any bare metal spots	DIY
☐ Install closed-cell spray foam insulation 2–3in all walls and ceiling	SUB
☐ Install 6mil vapor barrier and tape all seams	DIY
☐ Apply epoxy floor coating (320 sq ft) — 2 coats with cure time	EITHER
☐ Install FRP wall panels with adhesive — all zones	DIY
☐ Seal all penetrations (conduit, duct, pipe) before proceeding	DIY

PHASE 2 — ELECTRICAL ROUGH-IN

TASK	WHO
☐ Install 100A sub-panel in Z1 utility area	DIY
☐ Pull service wire from site utility panel — get permit	SUB
☐ Run conduit trunk lines full length of container overhead	DIY
☐ Pull branch circuit wire to all zones per single-line diagram	DIY
☐ Install GFCI outlets every 4ft along back wall (x12)	DIY
☐ Install light fixtures — T5 in Z3, CMH in Z4, HPS in Z5, shops in Z1/Z2/Z6	DIY
☐ Install all timers and smart outlets for grow light circuits	DIY
☐ Test all circuits with load — verify breakers, GFCI trip, continuity	DIY

PHASE 3 — HVAC & ENVIRONMENTAL SYSTEMS

TASK	WHO
☐ Install mini-split AC unit in Z1 — hire HVAC tech for refrigerant charge	SUB
☐ Wire mini-split low-voltage controls and line voltage supply	DIY
☐ Install inline exhaust fans x3 with backdraft dampers	DIY
☐ Install HEPA filter inline housings x2 (Z3 positive pressure supply)	DIY
☐ Run ducting — all exhaust and supply runs	DIY
☐ Install Inkbird humidity controllers — wire probes and relays	DIY
☐ Install ultrasonic humidifiers x2 in Z5 on dedicated circuit	DIY
☐ Install oscillating fans in Z4 and Z5	DIY
☐ Test all climate zones — verify temp and RH hit targets	DIY

PHASE 4 — PLUMBING & IRRIGATION

TASK	WHO
☐ Hire plumber — run site water supply line to container inlet	SUB
☐ Hire plumber — run drain line to site sewer or dry well	SUB
☐ Install utility sink in Z2 — connect supply and drain	SUB
☐ Install 5-stage RO system in Z1 — connect to supply line	DIY
☐ Install nutrient reservoir (55-gal) in Z1 — plumb to pump	DIY
☐ Run drip irrigation manifolds in Z4 and Z5 — connect to pumps	DIY
☐ Install Inkbird irrigation timers — set veg and flower schedules	DIY
☐ Pressure test all water connections — check for leaks 24hr	DIY

PHASE 5 — SHELVING, ZONES & FURNISHINGS

TASK	WHO
☐ Assemble NSF wire shelving units x5 — veg x2, fruiting x3	DIY
☐ Build propagation room partition — acrylic panels, door seal	DIY
☐ Install work counter (Z2) and harvest/trim table (Z6)	DIY
☐ Install mesh drying racks in Z6	DIY
☐ Install cold storage refrigerator in Z6 — verify temp	DIY
☐ Install blackout curtain at Z4/Z5 partition — test for light leaks	DIY
☐ Install all sensor hardware — mount, wire, commission	DIY
☐ Configure Home Assistant hub — verify all sensor reads	DIY

PHASE 6 — GRUSH INTEGRATION & COMMISSIONING

TASK	WHO
☐ Mount Grush tablet at Z3 and Z6 scan points	DIY
☐ Install label printer and test label output	DIY
☐ Configure Grush account — zones, shelf locations, batch types	DIY
☐ Print and apply QR labels to all shelf positions (Z4: 12, Z5: 18)	DIY
☐ Create test batch — run full workflow from propagation to harvest scan	DIY
☐ Verify environmental data logging to Grush batch records	DIY
☐ Run full systems test — all climate zones 48hr continuous	DIY
☐ Document any deviations from plan — update as-built notes	DIY
☐ Final walkthrough and sign-off	DIY

WHO KEY: DIY = Self-performed (no labor cost) SUB = Licensed trade required EITHER = Your call / time vs money